

(3)

Rockets

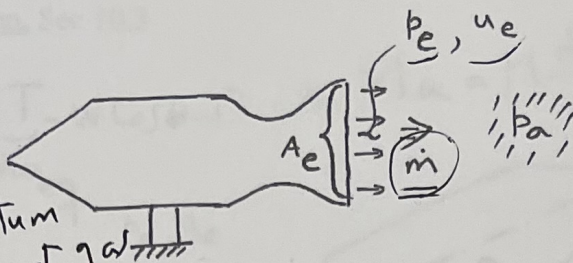


Diagram of a rocket nozzle. The nozzle is shown with an exit area A_e . The exhaust gas has pressure p_e and velocity u_e . The ambient pressure is p_a . The thrust force T is shown acting to the left. The momentum of the exhaust gas is indicated by a curved arrow pointing to the right.

Momentum of exhaust gas

$$\tau = \dot{m}u_e + (p_e - p_a)A_e \quad (10.4)$$

$$u_{eq} = \frac{\tau}{\dot{m}} = u_e + \frac{(p_e - p_a)A_e}{\dot{m}}$$

Since $\dot{m} = \rho_e u_e A_e$,

$$u_{eq} = u_e + \frac{(p_e - p_a)A_e}{\rho_e u_e A_e} = u_e + \frac{(p_e - p_a)}{\rho_e u_e}$$

Using the ideal gas equation, $p_e = \rho_e R T_e$

$$u_{eq} = u_e + \frac{RT_e}{u_e} \left(1 - \frac{p_a}{p_e}\right) \quad (\text{verify})$$

The specific impulse I_{sp} is

$$I_{sp} \stackrel{\text{def}}{=} \frac{u_{eq}}{g_e} = \frac{\tau}{\dot{m}g_e} \quad \{\text{thrust per unit weight flow rate of propellant}\} \quad g_e = 9.81 \text{ m/s}^2.$$

(see Eqn. 10.7)

Specific impulse of chemical rockets varies from 200 s to 450 s

Specific impulse of nuclear thermal rockets is about 900 s

Specific impulse of electrostatic thrusters is greater than 2000 s.

Example specific impulse values:

Falcon 9 (Space X): 280 s (SL), 350 s (vac)

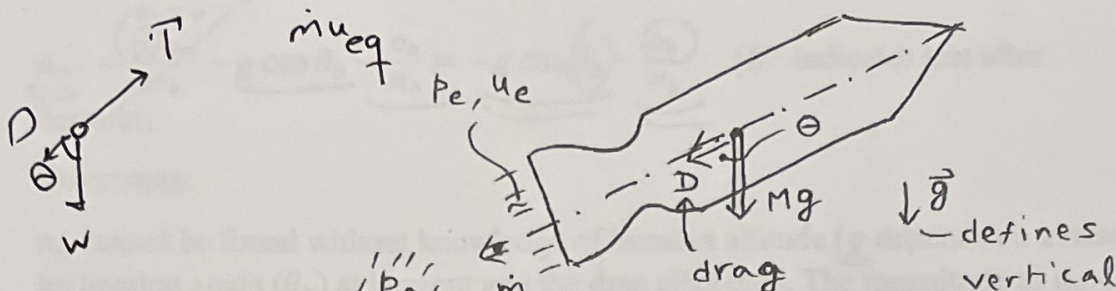
RD-170 (Atlas): 309 s (SL)

RS-25 (SSME): 453 s (vac)

RSRM: 267 s (SL)

Vehicle Acceleration, Sec 10.3

$$F = T - W \cos \theta - D = Ma = M \frac{du}{dt}$$



Mass

$$\frac{dM}{dt} = -\dot{m} \rightarrow M = M_0 - \dot{m}t$$

M_0 - initial mass (at $t = 0$), M - mass at time t

$$M_b = M_0 - \dot{m}t_b$$

M_b - burnout mass (at $t = t_b$), t_b - burn time

$$\text{Also note that } \dot{m}t_b = M_p \rightarrow M_b = M_0 - M_p$$

M_p - mass of propellant

$$\text{Mass ratio } R \stackrel{\text{def}}{=} \frac{M_0}{M_b}$$

$$R = \frac{M_0}{M_b}$$

Momentum equation

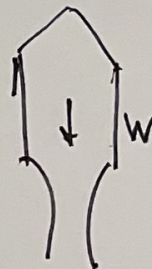
$$M \frac{du}{dt} = \dot{m}u_{eq} - Mg \cos \theta - D$$

$$\text{Acceleration: } a = \frac{du}{dt} = \frac{\dot{m}u_{eq}}{M} - g \cos \theta - \frac{D}{M}$$

$$\text{Launch acceleration: } a_0 = \frac{\dot{m}u_{eq}}{M_0} - \frac{g \cos \theta}{1} - \frac{D}{M_0} \quad \{\theta = 0 \text{ for vertical launch}\}$$

(typical)}

$$a_0 = \frac{\dot{m}u_{eq}}{M_0} - g_e$$



Burnout acceleration:

$$a_{b^-} = \frac{\dot{m} u_{eq}}{M_b} - g \cos \theta_b - \frac{D_b}{M_b} \quad \{b^- \text{ indicates just before burnout}\}$$

$$\cancel{a_{b^+} = \frac{\dot{m} u_{eq}}{M_b} - g \cos \theta_b - \frac{D_b}{M_b}} = -g \cos \theta_b - \frac{D_b}{M_b} \quad \{b^+ \text{ indicates just after burnout}\}$$

Comments:

a_b cannot be found without knowledge of burnout altitude (g depends on altitude), inclination angle (θ_b) at burnout and the drag (D_b) then. The magnitude of a_{b^-} , however, will be significantly larger than that of a_0 since the propellant mass is a large fraction of the rocket initial mass (91 % for the moon rocket, 86 % for the space shuttle, for example).

Short burn time results in high $\dot{m}$, which, in turn, results in large a_{b^-} . Thus t_b should not be too small.

Long burn time may result in insufficient a_0 , and high gravity- and drag-loss (discussed later).

Typical burn time for launch vehicle is on the order of several minutes {RSRM 2 minutes, SSME 8 minutes, moon rocket - 1.5 minutes (stage 1), 6 minutes (stage 2), 8 minutes (stage 3), Falcon 9 - 2.6 minutes (stage 1), 7 minutes (stage 2)}.

Velocity Rise

The function of a rocket is to provide velocity rise due to the payload. To find it, integrate the acceleration equation:

$$\frac{du}{dt} = \frac{\dot{m} u_{eq}}{M} - g \cos \theta - \frac{D}{M}$$

Since $\dot{m} = -\frac{dM}{dt}$,

$$\left[\frac{du}{dt} = -\frac{u_{eq}}{M} \frac{dM}{dt} - g \cos \theta - \frac{D}{M} \right] dt$$

$$\rightarrow du = -u_{eq} \frac{dM}{M} - g \cos \theta dt - \frac{D}{M} dt \quad (\text{see Eqn. 10.9}) \rightarrow \text{integrate}$$

$$\int_{u=u_0}^{u=u_b} du = - \int_{M_0}^{M_b} u_{eq} \frac{dM}{M} - \int_{t=0}^{t=t_b} g \cos \theta dt - \int_{t=0}^{t=t_b} \frac{D}{M} dt$$

$$u_b - u_0$$

$$\rightarrow (u_b - u_0) \stackrel{\text{def}}{=} \Delta u_b = \underbrace{u_{eq} \ln \left(\frac{M_0}{M_b} \right)}_{\Delta u_{b, \text{imp}}} - \underbrace{g \cos \theta}_{\text{gravity}} t_b - \underbrace{\left(\frac{D}{M} \right)}_{\text{drag}} t_b, \text{ in which}$$

$\overline{g \cos \theta} = \frac{\int_{t=0}^{t=t_b} g \cos \theta dt}{t_b}$, and $\overline{\left(\frac{D}{M} \right)} = \frac{\int_{t=0}^{t=t_b} \frac{D}{M} dt}{t_b}$, are just concise representations of the gravity and drag integrals, respectively.

Impulsive Velocity Rise ($\Delta u_{b, \text{imp}}$)

Impulsive velocity rise is defined to be the velocity rise attained with zero burn time ($t_b = 0$).

$$\Delta u_{b, \text{imp}} = u_{eq} \ln \left(\frac{M_0}{M_b} \right)$$

$$\Delta u_b = \Delta u_{b, \text{imp}} - \overline{g \cos \theta} t_b - \overline{\left(\frac{D}{M} \right)} t_b$$

It is clear that finite burn time decreases the attainable velocity rise. Longer the burn, greater are the gravity and drag losses (mentioned earlier).

Gravity

$$\frac{g}{g_e} = \left(\frac{R_e}{R_e + h} \right)^2 \quad (\text{Eqn. 10.12})$$

g - acceleration due to gravity at altitude h

g_e - acceleration due to gravity at earth's surface = 9.81 m/s^2

R_e - radius of the earth (6378 km)

Drag

$$D = C_D \frac{1}{2} \rho u^2 A_f \quad (\text{Eqn. 10.14}), \text{ in which}$$

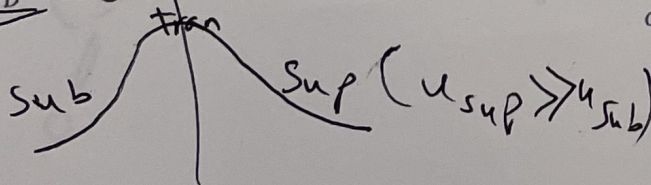
D - drag

$$\rho = a e^{-bh}$$

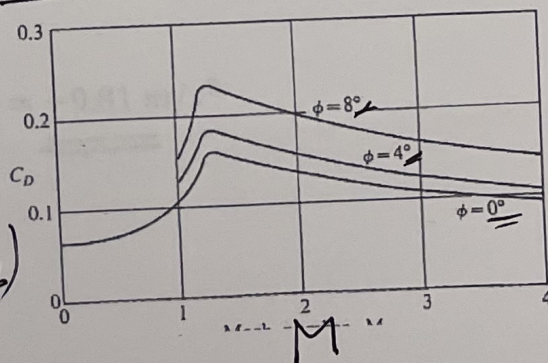
ρ - density $\{ \rho = a \exp(-bh^{1.15}), a = 1.2, b = 2.9(10^{-5}), h \text{ in m}, \rho \text{ in } \frac{\text{kg}}{\text{m}^3} \}$

A_f - frontal area

C_D - drag coefficient (See Fig. 10.4)

Sub  Sup ($u_{\text{sup}} \gg u_{\text{sub}}$)

$$\rho = 1.2 \text{ kg/m}^3$$



Example

Consider a rocket: $\dot{m} = 8740$ kg/s, $u_e = 2600$ m/s, $p_e = 71$ kPa, $A_e = 11.35$ m².

$M_0 = 1,720,000$ kg, $M_b = 680,000$ kg.

Vertical ascent ($\theta = 0$), $g = g_e$. Neglect drag.

Find τ_{SL} , $u_{eq,SL}$, $I_{sp,SL}$, $u_{eq,vac}$, $I_{sp,vac}$, a_0 , a_{b-} , a_{b+} , Δu_b

$$\tau = \dot{m}u_e + (p_e - p_a)A_e$$

$$p_a = 101 \text{ kPa at SL}$$

$$\tau_{SL} = \dot{m}u_e + (p_e - p_{a,SL})A_e$$

$$= (8740)(2600) + (71 - 101)(1000)(11.35) = 2.24 (10^7) \text{ N}$$

$$u_{eq,SL} = \frac{\tau_{SL}}{\dot{m}} = \frac{2.24 (10^7)}{8740} = 2563 \text{ m/s}$$

$$I_{sp,SL} = \frac{u_{eq,SL}}{g_e} = \frac{2563}{9.81} = 261 \text{ s}$$

$$p_a = 0 \text{ in vacuum.}$$

$$\tau_{vac} = \dot{m}u_e + (p_e - p_{a,vac})A_e$$

$$= (8740)(2600) + (71 - 0)(1000)(11.35) = 2.35 (10^7) \text{ N}$$

{Important: The values of p_e , u_e and $\dot{m}$ stay fixed for a rocket.}

$$u_{eq,vac} = \frac{\tau_{vac}}{\dot{m}} = \frac{2.35 (10^7)}{8740} = 2689 \text{ m/s}$$

$$I_{sp,vac} = \frac{u_{eq,vac}}{g_e} = \frac{2689}{9.81} = 274 \text{ s}$$

$$a_0 = \frac{\dot{m}u_{eq,SL}}{M_0} - g_e = (8740)(2563)/1,720,000 - 9.81 = 3.21 \text{ m/s}^2 = 0.33 g_e$$

$$a_{b-} = \frac{\dot{m}u_{eq,SL}}{M_b} - \underbrace{\tilde{g}}_{\tilde{g}} \underbrace{\cos \theta_b}_0 - \frac{\overset{0}{\tilde{D}}_b}{M_b} = (8740)(2563)/680,000 - 9.81 = 23.1 \text{ m/s}^2 = 2.36 g_e$$

$$a_{b+} = \frac{\overset{0}{\tilde{m}}u_{eq}}{M_b} - g \cos \theta_b - \frac{D_b}{M_b} = -\underbrace{\tilde{g}}_{\tilde{g}} \underbrace{\cos \theta_b}_0 - \frac{\overset{0}{\tilde{D}}_b}{M_b} = -9.81 \text{ m/s}^2$$

!

$$M_0 = M_b + \dot{m} t_b$$

$$\text{Burn time: } \underline{t_b} = \frac{M_0 - M_b}{\dot{m}} = \frac{1,720,000 - 680,000}{8740} = \underline{119 \text{ s}}$$

$$\underline{\Delta u_b} = u_{eq} \ln \left(\frac{M_0}{M_b} \right) - \int_{t=0}^{t=t_b} \cancel{g \cos \theta} dt - \int_{t=0}^{t=t_b} \frac{\cancel{D}}{\cancel{M}} dt$$

$$\{ \overline{g \cos \theta} = \frac{\int_{t=0}^{t=t_b} g \cos \theta dt}{t_b} = \frac{g_e t_b}{t_b} = \underline{g_e} \}$$

$$\Delta u_b = u_{eq} \ln \left(\frac{M_0}{M_b} \right) - \underline{g_e t_b}$$

$$= (2563) \ln (1,720,000/680,000) - (9.81)(119) = 2378 - 1167 = 1211 \text{ m/s} = 1.21 \text{ km/s.}$$

Observe the significant magnitude of the gravity loss. In a LEO launch, the gravity and drag losses can amount to 20% of the velocity rise the rocket is designed to provide.